

Qwen3.5-0.8B - ARC-Easy 5-case review

Greedy free generation - A113X-specialized C runtime - ThirdReality TRHub-V3 - measured 2026-08-12

Scope: This is a five-case generative smoke adaptation. It is not the official ARC-Easy likelihood evaluation and must not be cited as an ARC-Easy benchmark score. Primary timing and resource measurements are from the A113X target.

4 / 5

exact generated answers

2.9182 tok/s

A113X aggregate prompt prefill

1.8220 tok/s

A113X aggregate steady decode

333.8% / 490 MiB

weighted CPU / maximum RSS

Method

Dataset	AI2 Reasoning Challenge, ARC-Easy validation split
Selection	First five source-order rows whose labels are exactly A, B, C, D. Selection was fixed before inference.
Input	Five distinct runtime strings passed through --prompt. None is compiled into the image.
Generation	Greedy; maximum 32 tokens; all five stopped at EOS after 9 tokens.
Scoring fixed before run	Extract the last case-sensitive standalone Answer: [A-D]. Missing or ambiguous output is incorrect.
Source pin	Canonical revision 7f5ff0bd482eee246ed06220b2450de1553890c6; validation parquet SHA-256 4e3752de16e4255ba97c6d521435fab222034acb129636b99569eefcd43ff0a0.
Image	492,962,304 bytes; SHA-256 00bb5a62214f1db44722731f895eeb9d9d28a27b0201bdd9287485671e5fd474

Measured command shape

```
prompt=$(jq -r '.cases[0].prompt' profile.json)
OMP_NUM_THREADS=4 /usr/bin/time -v ./qwen35-task-a113x-generate qwen35-v2.qtask --prompt "$prompt" --generate 32
```

A113X result table

# / case	Ref	Out	Result	Tokens	Prefill	TTFT	Decode	Wall	CPU	Peak RSS	Swap
1 / MCAS_2000_4_6	A	B	wrong	69	22.561309 s	22.671961 s	1.858670 tok/s	27.05 s	345%	500,864 KiB	0
2 / Mercury_7057260	C	C	correct	106	37.758794 s	37.869394 s	1.850583 tok/s	42.27 s	325%	502,016 KiB	0
3 / ACTAAP_2014_7_6	A	A	correct	93	29.735323 s	29.845795 s	1.969770 tok/s	33.98 s	354%	501,632 KiB	0
4 / Mercury_7122448	B	B	correct	101	37.065897 s	37.176777 s	1.728581 tok/s	41.88 s	316%	501,760 KiB	0
5 / Mercury_SC_416516	B	B	correct	97	32.568104 s	32.679591 s	1.724987 tok/s	37.39 s	335%	501,120 KiB	0

Aggregate timing and resources

Prompt prefill	466 tokens / 159.689427 s	2.918164 tokens/s
Mean time to first token	five invocations	32.048704 s
Steady decode	40 tokens / 21.954294 s	1.821967 tokens/s
Generation after prefill	45 tokens / 22.508385 s	1.999255 tokens/s
Total process wall	five separate invocations	182.57 s
CPU	605.76 s user + 3.74 s system	333.84% wall-weighted
Peak memory	maximum process RSS	502,016 KiB (490.25 MiB)
Swap	process and whole-device checks	0 before, during, and after
Page faults	0 major	91,102 minor
Context switches	10,733 voluntary	208,531 involuntary

Runtime comparison

Runtime / CPU	Exact	Prefill	Steady decode	Mean TTFT	Wall	Max RSS	Swap
A113X-specialized C / 4x Cortex-A53	4/5	2.9182 tok/s	1.8220 tok/s	32.0487 s	182.57 s	502,016 KiB	0
Generic C / Apple M3 Pro	4/5	3.7094 tok/s	2.3869 tok/s	25.3308 s	143.46 s	514,867,200 B	0

Case 1: MCAS_2000_4_6

Reference: A	Generated: B	WRONG
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Exact runtime input

```
<|im_start|>system
Answer the multiple-choice science question. Do not explain. End with exactly one line: Answer: X, where X is A, B, C, or
D.<|im_end|>
<|im_start|>user
Which technology was developed most recently?
A. cellular telephone
B. television
C. refrigerator
D. airplane<|im_end|>
<|im_start|>assistant
```

Exact human-readable model output

```
<think>

</think>

Answer: B
```

Prompt tokens	69	Prefill	22.561309 s / 3.058333 tok/s
TTFT	22.671961 s	Steady decode	4.304153 s / 1.858670 tok/s
Wall	27.05 s	CPU	345% (92.98 s user + 0.62 s system)
Peak RSS	500,864 KiB	Swap	0
Page faults	0 major / 18,070 minor	Context switches	1,468 voluntary / 32,375 involuntary

Case 2: Mercury_7057260

Reference: C	Generated: C	CORRECT
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Exact runtime input

```
<|im_start|>system
Answer the multiple-choice science question. Do not explain. End with exactly one line: Answer: X, where X is A, B, C, or
D.<|im_end|>
<|im_start|>user
A student hypothesizes that algae are producers. Which question will best help the student determine if this is correct?
A. Do algae consume other organisms?
B. Which organisms consume algae?
C. Do algae use sunlight to make food?
D. Could an ecosystem survive without algae?<|im_end|>
<|im_start|>assistant
```

Exact human-readable model output

```
<think>
</think>
Answer: C
```

Prompt tokens	106	Prefill	37.758794 s / 2.807293 tok/s
TTFT	37.869394 s	Steady decode	4.322962 s / 1.850583 tok/s
Wall	42.27 s	CPU	325% (136.88 s user + 0.85 s system)
Peak RSS	502,016 KiB	Swap	0
Page faults	0 major / 18,306 minor	Context switches	3,394 voluntary / 48,241 involuntary

Case 3: ACTAAP_2014_7_6

Reference: A	Generated: A	CORRECT
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Exact runtime input

```
<|im_start|>system
Answer the multiple-choice science question. Do not explain. End with exactly one line: Answer: X, where X is A, B, C, or
D.<|im_end|>
<|im_start|>user
Soccer players use their muscle systems to kick a ball into a goal. What organ system coordinates the muscles?
A. The nervous system
B. The endocrine system
C. The respiratory system
D. The circulatory system<|im_end|>
<|im_start|>assistant
```

Exact human-readable model output

```
<think>

</think>

Answer: A
```

Prompt tokens	93	Prefill	29.735323 s / 3.127593 tok/s
TTFT	29.845795 s	Steady decode	4.061389 s / 1.969770 tok/s
Wall	33.98 s	CPU	354% (119.58 s user + 0.78 s system)
Peak RSS	501,632 KiB	Swap	0
Page faults	0 major / 18,219 minor	Context switches	2,002 voluntary / 42,767 involuntary

Case 4: Mercury_7122448

Reference: B	Generated: B	CORRECT
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Exact runtime input

<|im_start|>system
Answer the multiple-choice science question. Do not explain. End with exactly one line: Answer: X, where X is A, B, C, or D.<|im_end|>
<|im_start|>user
Planets in the solar system are in constant motion. What factor has the greatest effect on the orbits of the planets?
A. the size of the planets
B. gravitational pull of the Sun
C. the composition of the planets
D. electromagnetic radiation from the Sun<|im_end|>
<|im_start|>assistant

Exact human-readable model output

<think>

</think>

Answer: B

Prompt tokens	101	Prefill	37.065897 s / 2.724877 tok/s
TTFT	37.176777 s	Steady decode	4.628073 s / 1.728581 tok/s
Wall	41.88 s	CPU	316% (131.77 s user + 0.76 s system)
Peak RSS	501,760 KiB	Swap	0
Page faults	0 major / 18,271 minor	Context switches	2,033 voluntary / 44,740 involuntary

Case 5: Mercury_SC_416516

Reference: B	Generated: B	CORRECT
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Exact runtime input

```
<|im_start|>system
Answer the multiple-choice science question. Do not explain. End with exactly one line: Answer: X, where X is A, B, C, or
D.<|im_end|>
<|im_start|>user
How is a pond different from a lake?
A. Ponds have moving water.
B. Ponds are smaller and shallower.
C. Ponds are not surrounded by land.
D. Ponds have a different amount of salt.<|im_end|>
<|im_start|>assistant
```

Exact human-readable model output

```
<think>

</think>

Answer: B
```

Prompt tokens	97	Prefill	32.568104 s / 2.978374 tok/s
TTFT	32.679591 s	Steady decode	4.637717 s / 1.724987 tok/s
Wall	37.39 s	CPU	335% (124.55 s user + 0.73 s system)
Peak RSS	501,120 KiB	Swap	0
Page faults	0 major / 18,236 minor	Context switches	1,836 voluntary / 40,408 involuntary

Audit data and references

Profile SHA-256	28d250832265c69c6dd81b0beec94c8eafefed2ea2e5c1bd7776ab6ebc95e05f
A113X binary SHA-256	c3a12f6b702a1bac7b91884f4e4ada189d53c91ef1ad551153dfb9a2a9fefe5a
Image SHA-256	00bb5a62214f1db44722731f895eeb9d9d28a27b0201bdd9287485671e5fd474
Canonical dataset	https://huggingface.co/datasets/allenai/ai2_arc/tree/7f5ff0bd482eee246ed06220b2450de1553890c6/ARC-Easy
ARC paper	https://arxiv.org/abs/1803.05457
Standard harness task	https://github.com/EleutherAI/lm-evaluation-harness/tree/main/lm_eval/tasks/arc

Generated token IDs

1 / MCAS 2000 4 6: [198, 248068, 271, 248069, 271, 15666, 25, 417, 248046]
2 / Mercury 7057260: [198, 248068, 271, 248069, 271, 15666, 25, 351, 248046]
3 / ACTAAP_2014_7_6: [198, 248068, 271, 248069, 271, 15666, 25, 357, 248046]
4 / Mercury 7122448: [198, 248068, 271, 248069, 271, 15666, 25, 417, 248046]
5 / Mercury_SC_416516: [198, 248068, 271, 248069, 271, 15666, 25, 417, 248046]